

KONRAD KOCKLER

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RESEARCH EXPERIENCE

PhD research, ITP Heidelberg

since 2026

- PhD researcher in the group "Strongly Correlated Systems" under Jan Pawłowski
- Investigating the QCD phase diagram at high baryon density using functional renormalization group methods
- Development of a momentum-dependent truncation combining a quark–gluon sector at high scales with a dynamically hadronised quark–meson sector at low scales
- Accessing non-perturbative phenomena including the critical endpoint, moat regime, and inhomogeneous phases
- Deriving experimentally testable predictions for the CBM experiment at FAIR

Bachelor and Master work, ITP Heidelberg

2023 - 2025

- First self-consistent, manifestly Lorentz-invariant computation of spectral functions in the Quark-Meson model
- Research on spectral functions and real-time dynamics
- Scaling of the spectral function of 3d φ^4 theory close to the phase transition
- Spectral functions of the Quark-Meson model both in vacuum and at finite temperature

PUBLICATIONS

- [1] Konrad Kockler, Jan M. Pawłowski, Franz R. Sattler, Ruwen Schulz, and Jonas Wessely. "Multi-scattering processes and spectral properties of low-energy QCD". In: (Aug. 2026). arXiv: [2608.18853 \[hep-ph\]](#).
- [2] Konrad Kockler, Jan M. Pawłowski, and Jonas Wessely. "Critical scaling for spectral functions". In: *Eur. Phys. J. C* 85.9 (2025), p. 950. DOI: [10.1140/epjc/s10052-025-14679-9](#). arXiv: [2506.09142 \[hep-th\]](#).

RELEVANT RESEARCH SKILLS

Theoretical Methods

- Use of functional Renormalisation Group techniques for non-perturbative QFT
- Spectral function techniques in Minkowskian spacetime

Computational Skills

- Numerical solution of flow equations
- Inclusion both full (euclidean) momentum dependence and full effective potential
- Real-time dynamics in quantum field theory, full self-consistent inclusion of higher order scattering contributions

Programming

- Extensive use of Mathematica for the symbolical generation of flow equations
- Resulting integral-differential equations are solved numerically using Julia and C++

EDUCATION

University of Heidelberg

PhD in Theoretical Physics

since 2026

University of Heidelberg

M.Sc. Physics

2023-2025

Master thesis "*Self-consistent spectral functions of the Quark-Meson model at finite temperatures*"

Grade: 1.0

University of Heidelberg

B.Sc. Physics

2020 - 2023

Bachelor thesis "*Into the scaling limit of scalar ϕ^4 theory: A spectral fRG approach*"

Grade: 1.1

WORK HISTORY AND INTERNSHIPS

Member of RF2.0 project at DESY

since 2024

- Development of methods to reduce CO₂ emissions at datacenters with minimal impact on users
- Analysis of the carbon intensity of German electricity mix
- Quantifying the makeup and usage of the DESY datacenter, to use as model for simulating a digital twin for testing of energy saving policies

DESY summer school

16.07.24-05.09.24

- Student internship at the IT group at DESY in Hamburg
- Work in the "Research facility 2.0" EU project

Tutor and Head Tutor in Theoretical Physics

2023-2026

- Led weekly tutorial sessions, reinforcing lecture material and guiding students through exercises.
- Tutor: Theoretical Electrodynamics (WS 2023/24); Higher Mathematics for Physicists (SS 2024, WS 2024/25); Quantum Mechanics (SS 2025); and Introduction to the Functional Renormalisation Group (SS 2026).
- Head Tutor: Quantum Field Theory I (WS 2026/27).